Biomedical Signal Processing and Control 86 (2023) 105330

Contents lists available at ScienceDirect

Biomedical Signal Processing and Control

journal homepage: www.elsevier.com/locate/bspc

MMS-Net: Multi-level multi-scale feature extraction network for medical image segmentation

Chang Zhao [a,1], Wenbing Lv [b,1], Xiang Zhang [a], Zimin Yu [a], Shunfang Wang [a,c,*]

[a] Department of Computer Science and Engineering, School of Information Science and Engineering, Yunnan University, Kunming, 650504, Yunnan, China
[b] Department of Electronic Engineering, School of Information Science and Engineering, Yunnan University, Kunming, 650504, Yunnan, China
[c] The Key Lab of Intelligent Systems and Computing of Yunnan University, Kunming, 650504, Yunnan, China

ARTICLE INFO

Keywords:
Automatic segmentation
Skip connection modules
Global multi-scale feature
Feature pyramid network
Local multi-scale context

ABSTRACT

Automatic segmentation of medical images can provide a reliable basis for clinical analysis and disease diagnosis. Due to the varying sizes and appearances of different pathological lesions, most convolutional neural networks (CNNs) approaches are unable to fully extract multi-scale features and usually lead to false positive and false negative segmentation. Multi-scale medical image segmentation still faces significant challenges. By designing different levels of skip connection modules (SCMs), local multi-scale context is proposed. We also adopted feature pyramid network (FPN) to connect SCMs to extract global multi-scale features. Local multi-scale context and global multi-scale feature constitute a multi-level multi-scale feature extraction network (MMS-Net). Multi-level multi-scale feature was extracted and accurate spatial information was also retained. Comprehensive experiments (training vs. testing: 8:2) on five public datasets show that the proposed MMS-Net achieved accurate segmentation and outperformed the state-of-the-art algorithms. Besides, when only using 50% and 20% of the training dataset, MMS-Net still obtained competitive segmentation results.

1. Introduction

Medical image segmentation is an essential intermediate step towards subsequent quantitative evaluation, diagnosis, staging, therapy planning, prognosis prediction, and treatment decision-making. Conventional manual segmentation is highly dependent on the experience of clinicians, labor extensive, and usually suffers from inter-observer variations. Automatic accurate lesion or organ segmentation is thus urgently demanded in various kinds of clinical practice, e.g., lung segmentation in CT images [1,2], polyp segmentation in endoscopy [3,4], dermatosis segmentation [5,6] and chest segmentation in X-ray images [7–10]. As shown in Fig. 1, differences in image modality, disease type and individual will affect the accuracy of final segmentation results.

When using U-Net to extract multi-scale features, the up-sample operation in the decoder will restore the feature image to the original scale, but this up-sample method is insufficient to achieve detail recovery, so some details may be lost during multi-scale feature extraction, resulting in inaccurate segmentation results. On the premise of ensuring the reliability of data and gradient, only four layers of encoder–decoder were retained in U-Net++, which limits the ability of feature extraction. We thus proposed a multi-level and multi-scale feature extraction network (MMS-Net) for medical image segmentation. MMS-Net is an encoder–decoder structure that can simultaneously extract and aggregate high-level semantic features and low-level detail features. Specifically, at each level of the encoding sub-network, we use the skip connection module to extract local multi-scale context and use the feature pyramid network to cascade different SCMs to obtain global multi-scale features. Local multi-scale context and global multi-scale features constitute multi-level multi-scale feature extraction. In the down-sample stage, we add multiple multi-level multi-scale feature extraction modules to enhance the semantic information of the extracted multi-scale features. In the up-sample stage, we use the same encoder structure to restore the position information of the input image. We combine the extracted semantic information representing pathological features with location information through lateral connection. Finally, we get more abundant multi-scale features through encoding and decoding and get the final segmentation prediction results. We use SCMs to deepen the network layers without generating gradient problems, which ensures that the extracted features are more representative, and better performance can be achieved even with limited training data.

* Corresponding author at: Department of Computer Science and Engineering, School of Information Science and Engineering, Yunnan University, Kunming, 650504, Yunnan, China.
E-mail address: sfwang_66@ynu.edu.cn (S. Wang).
[1] These authors share first authorship.

https://doi.org/10.1016/j.bspc.2023.105330
Received 29 March 2023; Received in revised form 9 July 2023; Accepted 1 August 2023
Available online 18 August 2023
1746-8094/© 2023 Elsevier Ltd. All rights reserved.

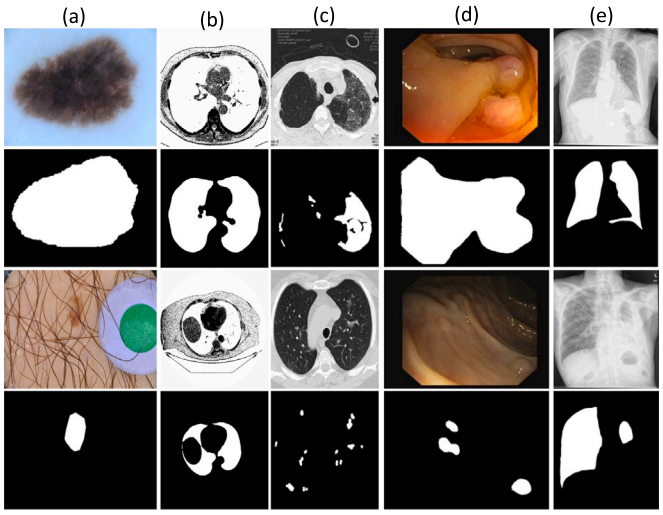


Fig. 1. Example of representative medical images and corresponding ground truths. (a) Dermatosis images from ISIC2017 dataset for melanoma segmentation, (b) Lung CT images from LUNA16 dataset for lung field segmentation, (c) Lung CT images from COVID-19 dataset for pneumonia segmentation, (d) Endoscopy images from CVC-ClinicDB dataset for colorectal polyp segmentation, and (e) Chest X-ray images from ShenZhen dataset for lung field segmentation.

In summary, the contributions of this work are four-fold:

Firstly, we propose a module with the SCMs to extract the local multi-scale context. Different layers of SCMs are designed in different layers of MMS-Net to effectively ensure the relevance of feature extraction.

Secondly, we use the structural form of FPN to connect SCMs at different layers. with the advantage of mixing the semantic feature information and the location information to obtain the global multi-scale feature.

Thirdly, the structure of the MMS-Net is composed of multiple SCMs for extracting local multi-scale context and an FPN structure for extracting global multi-scale features. Local multi-scale context extraction and global multi-scale feature extraction constitute multi-level multi-scale feature extraction, enabling the superior performance of MMS-Net compared with other classical models.

Finally, we conducted comprehensive experiments and achieved excellent state-of-the-art segmentation performance (80% for training, 20% for testing) in five different datasets. In addition, to verify the effectiveness of MMS-Net in the scenario of limited training datasets, we reduced the training datasets to 50% and 20%. The results showed that MMS-Net still performs well regardless of the decreased number of training data and deepened number of layers in MMS-Net.

The rest of this paper is organized as follows: In the second section, we reviewed the development of multi-scale feature extraction methods and their applications in medical images. In the third section, we introduced our proposed multi-level multi-scale feature extraction network in detail. In the fourth section, we presented the implementation of experiments. In the fifth section, we evaluated the proposed model and compared it with the latest methods on five public datasets, results, discussions, and limitations are given. Finally, the sixth section summarized this paper and gave a conclusion.

2. Related works

2.1. Multi-scale feature extraction method

Multi-scale feature extraction methods have been widely studied in various computer vision tasks, while the down-sample in semantic segmentation will reduce the resolution of the image and further lead to the loss of image feature information. The dilated convolution with expansion rate was introduced, through the adjustment of the magnification factor, the convolution kernel with the same size has a larger receptive field [11]. DeeplabV3 replaces the last several blocks of ResNet with dilated convolution, which has the advantage of obtaining more intensive feature responses without reducing the resolution, thus making the position information restored by up-sample more accurate [12]. The image pyramid obtains the low-resolution image by down-sample the input image through the Gaussian kernel convolution recovers the high-resolution image by up-sample, and reconstructs the upper layer image through the Laplace transform, and the input images with different resolutions are aggregated to obtain multi-scale features. With the increase of the number of layers in the traditional top-down convolution network, the features of small-size images will gradually be lost, resulting in the degradation of detection performance. Feature fusion adds deep semantic information to the shallow network feature map through down-sample, up-sample, and lateral connections, and the semantic information of down-sample and the location information of up-sample are integrated, which improves the performance of segmentation of small objects [13]. In order to avoid the gradient problem caused by depth, the parallel multi-branch convolution structure was proposed to increase the width, specifically, multi-scale feature extraction and fusion are performed by splicing convolution kernels of different sizes [14,15]. Li et al. [2] proposed the multi-branches feature attention (MBFA) module. It consists of four branch convolution kernels, each with a different convolution kernel size. In this way, each branch has a different size of receptive field, which helps self-regulate to cope with multi-scale segmentation. In the study of Iqbal et al. [8], convolution, activation, and batch normalization operations are replaced with the dilated fusion block (DF). The major contribution of each DF is to adaptively adjust the network's receptive field and mix feature maps determined using different convolutions operations and effectively address the multi-scale challenges of lung images. Similar to Li et al. [2], Iqbal et al. [8] parallelized convolutional kernels of different branches to obtain feature maps of different sizes.

2.2. Application to medical image segmentation

It is well known that there are many multi-scale feature extraction methods have been applied to medical image segmentation. Qu et al. [16] proposed a multi-scale multi-view segmentation framework for accurate multi-phase pancreas segmentation. Geng et al. [17] proposed a refocusing method for cervical cytopathology images via multi-scale attention features and domain normalization. Valvano et al. [18] proposed a novel attention-gating mechanism, which they condition with adversarial signals to act as a shape prior, resulting in better object localization at multi scales. Motivated by memory-based decision-making and visual attention mechanism as a filter to select environmental information in the human visual perceptual system, Chen et al. [19] proposed a multi-scale attention memory with hash addressing auto-encoder network (MAMA-Net) for anomaly detection. Pulmonary fibrosis net (PF-Net) proposed by Wang et al. [20] combines 2D and 3D convolutions to deal with CT volumes with large inter-slice spacing, and multi-scale guided dense attention is used to segment complex pulmonary fibrosis lesions. Surgi-Net [21] was proposed to learn pyramid attention features, and the double attention module is designed to capture the semantic dependencies between locations and channels. Based on semantic dependencies, the semantic features in the disturbed area can be inferred for addressing illumination issues. Pyramid attention is aggregated to capture multi-scale features and make predictions more accurate. Dong et al. [22] proposed a scale-aware feature enhancement module, which aims to dynamically adjust the receptive field to effectively extract more expressive features, thus enhancing the network's feature representation ability.

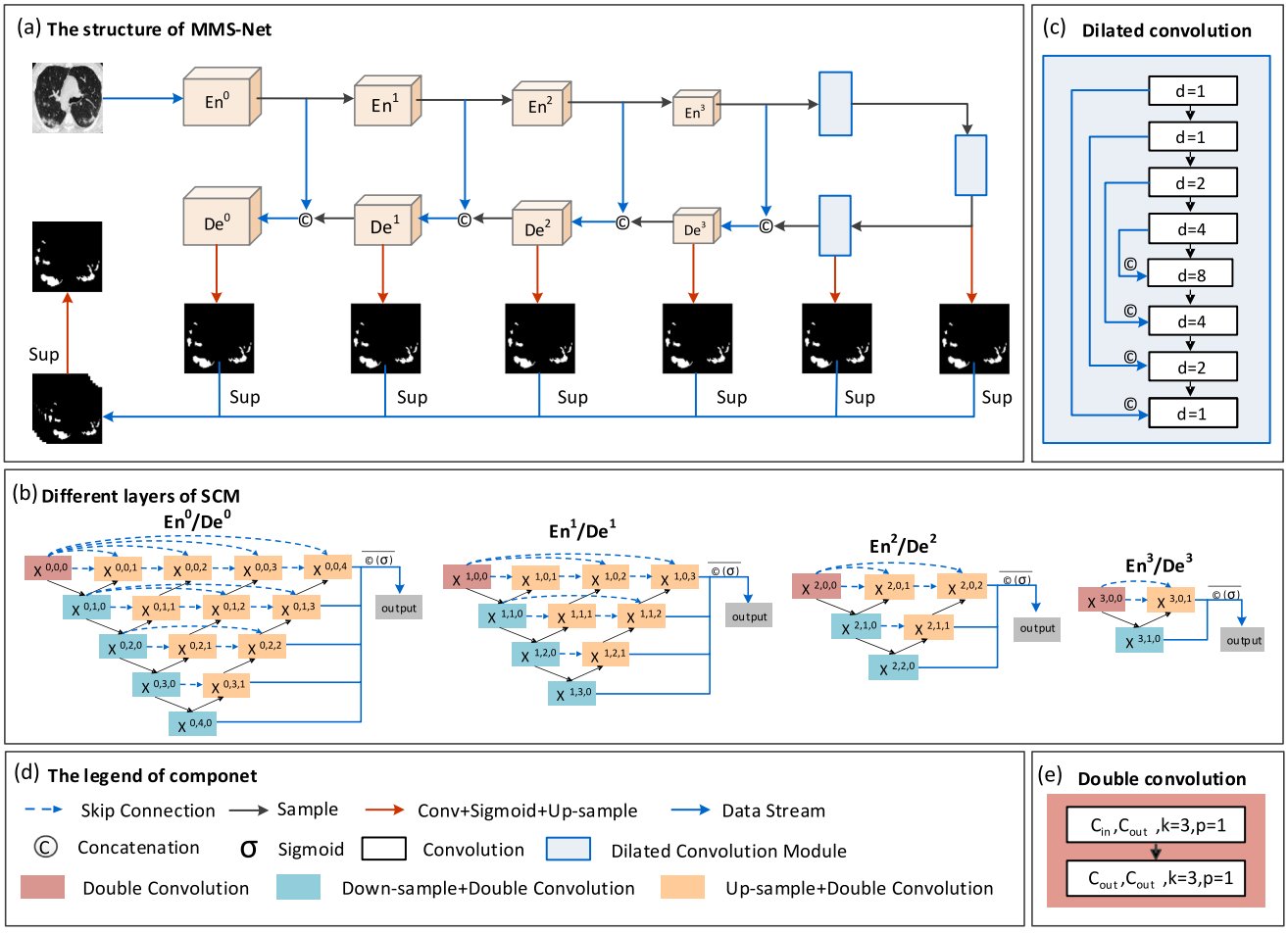


Fig. 2. Structure of the proposed multi-level multi-scale feature extraction network (MMS-Net). (a) The structure of MMS-Net, (b) Different layers of SCMs, (c) Dilated convolution, (d) The legend of component, and (e) Double convolution.

3. Proposed method

3.1. Overall architecture

Fig. 2 illustrates the overall architecture of MMS-Net, where a classic U-Net-based encoder–decoder structure was adopted as the backbone. Fig. 2(b) shows different layers of SCMs designed to extract local multi-scale context. As is shown in Fig. 2(a), between different encoder–decoder stages, three dilated convolution modules (Fig. 2(c)) are designed to ensure the expansion of the receptive field without reducing spatial resolution. It should be noted that the number of input and output channels of the second convolution in double convolution (Fig. 2(e)) is consistent, which helps MMS-Net learn the correlation between input and output and ensures to effectively extract the information from the input data. Finally, we introduce FPN to connect SCMs at different stages to extract global multi-scale features. Local multi-scale context and global multi-scale features constitute a multi-level multi-scale feature extraction method. Sup means deep supervision similar to holistically-nested edge (HED) [23].

3.2. Skip connection module

Although the residual U-block (RSU) module of U^2-Net can extract the global multi-scale feature on the substructure, it ignores the fusion of contexts between different levels of convolutions. Hence, inspired by U-Net++, we proposed SCMs to fuse features between the adjacent stages, features with different sizes were processed through down- or up-sample operation. Therefore, the diverse local multi-scale context can be effectively extracted by SCMs. Furthermore, SCMs with different layers can prevent gradient disappearance or explosion. Our SCMs differ from U-Net++ in some aspects. First, SCMs connect the outputs of each stage of the decoder and averages them to obtain the final result. Besides, considering the impact of layers on performance, MMS-Net only uses the accurate mode of U-Net++ and discards its fast mode. The accurate mode of U-Net++ calculates the average value of the output of all segmented branches. The fast mode of U-Net++ selects one of the branch outputs as the output, which determines the pruning degree and speed gain of the model. Since segmentation accuracy is more important than speed, especially for the analysis of medical images, thus we chose the accurate mode to obtain better segmentation performance

As shown in Fig. 2(b), SCMs consist of down-sample, up-sample, and double convolution. In double convolution and dilated convolution (Fig. 2(c, e)), C represents the channel of convolution, k represents the size of convolution kernel, p represents the number of boundary fills, and d represents the expansion factor.

It is known that the more layers of the deep learning network, the greater the possibility of gradient disappearance and explosion. Therefore, the deeper the MMS-Net, the SCMs with less number of layers. Specifically, we take into account the number of layers of SCMs, if a five-layer SCM structure is used in each encoder–decoder block of MMS-Net, even if the skip connection is used to relieve the gradient problem to a certain extent, the gradient problem will still occur in the overall model. So we reduce the layers of SCM and designed four different structures, i.e., five-, four-, three-, and two-layer SCM

respectively. The calculation formula of the concatenation layer and accurate mode is shown in Eq. (1).

$$x^{k,i,j} = \begin{cases} H\left(D\left(x^{k,i-1,j}\right)\right), j = 0 \\ H\left(\left[H\left(x^{k,i,p}\right)\Big|_{p=0}^{j-1}, U\left(H\left(x^{k,i+1,j-1}\right)\right)\right]\right), j > 0 \end{cases} \tag{1}$$

where $x^{k,i,j}$ represents the location of the cell in SCMs at different levels, k represents the number of layers in SCMs, i indexes the down-sample layer along the encoder and j indexes the convolution layer of the dense block along the skip pathway, H represents double convolution, and D represents down-sample. The range of i and j is $[0, 4 - k]$.

In short, we proposed SCMs to extract local multi-scale context, it can overcome the shortcomings of RSU. Local multi-scale context proposed in MMS-Net is extra by SCMs.

3.3. Feature pyramid network

Several strategies to connect SCMs of different layers for multi-scale feature extraction are shown in Fig. 3. As shown in Fig. 3(a), featurized image pyramid is time-consuming and does not have much improvement in performance. In Fig. 3(b), the output features are obtained from shallow layers and are not robust. For the strategy of pyramidal feature hierarchy as shown in Fig. 3(c), the semantic feature information of the input image is extracted by simple down-sample, and the down-sample sacrifices the resolution of the image, the resolution of the image can be restored by up-sample, but the lack of location information will cause the low performance of the model. The three strategies mentioned above all have shortcomings. Therefore, we use the structure of feature pyramid network (FPN) in Fig. 3(d) to connect SCMs of different layers. SCM is divided into encoder and decoder. Encoder and decoder are connected by down-sample and up-sample. The rule of their connection is shown in Eq. (2).

$$\begin{cases} En^k = D\left(En^{k-1}\right) \\ De^{k-1} = En^{k-1} \copyright \left(U(De^k)\right) \end{cases} \tag{2}$$

where En^k and De^k represent encoder and decoder respectively. En^0 represents encoder of four-layer encoder–decoder, En^1 represents encoder of three-layer encoder–decoder, De^0 represents decoder of four-layer encoder–decoder, and so on. The FPN used in our proposed MMS-Net is different from the traditional FPN, the output data of each decoder will be changed to the same size as the input data by up-sample instead of ROI pooling [24]. FPN adds an inter-layer fusion to the structure of pyramidal feature hierarchy. Through pooling between different layers of SCMs, feature maps of different scales are extracted. The semantic information of the image extracted by down-sample is fused with the position information extracted by up-sample through concatenation and determined a top-down, bottom-up, inter-layer integration structure. In order to facilitate the splicing of feature maps output by SCMs at different levels, we resize the output feature map of SCMs to the input image. This is the global multi-scale feature extraction strategy by using FPN. The global multi-scale feature and local multi-scale context constitute the multi-level multi-scale feature extraction method.

4. Experiments

4.1. Code and dataset availability

For the repeatability of the experiment, the code was made available at https://github.com/Zcjiayouya/MMSNet.

The proposed MMS-Net was used on 5 different modality datasets for the segmentation of different tissue or lesion as shown in Table 1. (1) 2000 dermatoscopy images from the 2017 melanoma detection dermatosis analysis challenge of the International Skin Imaging Co-operation (ISIC) [25], denoted as ISIC2017. (2) 267 CT images from LUNA 16 challenge for lung field segmentation [26]. (3) 100 axial CT images of 60 patients with COVID-19 from Italian Society of Medicine and Interventional Radiology for pneumonia segmentation [27] (including ground glass opacity, consolidation, and hydrothorax). (4) 612 endoscopy images in CVC-ClinicDB dataset from MICCAI 2015 sub-challenge on automatic colorectal polyp segmentation. Images are from Hospital Clinic, Barcelona and masks are property of Computer Vision Center, Barcelona, Spain [28]. (5) 453 chest X-ray images from ShenZhen No. 3 Hospital in ShenZhen, Guangdong, China, masks were prepared manually by students and teachers in the Department of Computer Engineering, and Faculty of Informatics and Computer Engineering, National Technical University of Ukraine, Kyiv, Ukraine [29].

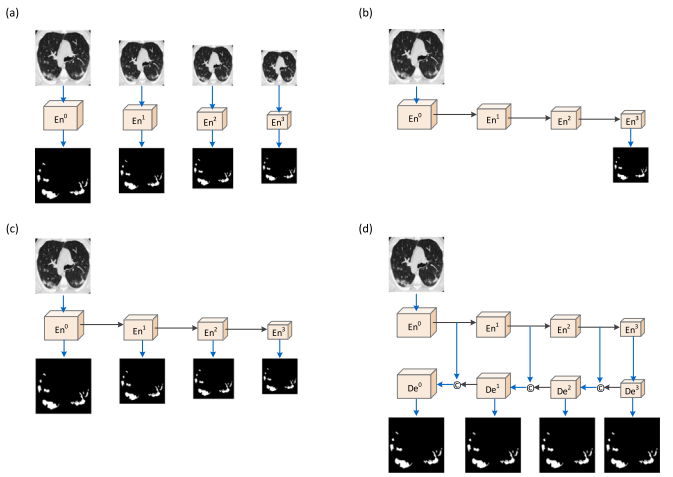


Fig. 3. Strategies to connect SCMs of different layers. (a) Featurized image pyramid, (b) Single feature map, (c) Pyramidal feature hierarchy, and (d) Feature pyramid network (FPN).

Table 1
A summary of the 5 datasets used in this study.

Dataset	Modality	Quantity	Image size
ISIC2017	Dermatoscope	2000	512-6728*512-4459
LUNA16	CT	267	512*512
COVID-19	CT	100	512*512
CVC-ClinicDB	Colonoscope	612	384*288
ShenZhen	X-ray	453	1233-3001*989-3001

4.2. Data pre-processing

Intensity normalization, size scaling and data augmentation were implemented in pre-processing step. Since large intensity value of the network input may result in saturation phenomenon of the network output, all images were first projected from [0, 255] to [0, 1] by dividing each gray level by 255 and then were normalized. For CT and X-ray images, single-channel intensity normalization was adopted following Eq. (3).

$$I_1 = \frac{I_0 - 0.485}{0.229} \tag{3}$$

where I_0 and I_1 are the intensity values before and after normalization, respectively. For dermatoscopy and endoscope images with RGB channels, three-channel intensity normalization was used following Eq. (4).

$$\begin{cases} I_1^R = \dfrac{I_0^R - 0.485}{0.229} \\ I_1^G = \dfrac{I_0^G - 0.456}{0.224} \\ I_1^B = \dfrac{I_0^B - 0.406}{0.225} \end{cases} \tag{4}$$

where I_0^R, I_0^G and I_0^B are the intensity values before normalization of R-, G-, and B-channel, respectively, while I_1^R, I_1^R and I_1^R are the intensity values after normalization of R-, G-, and B-channel, respectively.

Considering the heterogeneous image sizes among different datasets, we uniformly scale the size of the input image and its corresponding mask to 320*320 by bilinear interpolation. Data augmentation through random clipping was also implemented to improve the robustness of models as it can increase the number of training datasets. Specifically, all images are randomly flipped vertically and cropped to patches with the size of 288 * 288 while preserving image semantic information.

4.3. Model training

We divided each dataset into training and testing by a ratio of 8:2 for a fair comparison with the classical/baseline model and other state-of-the-art models. Adam is selected as the optimizer of MMS-Net. Specific parameter settings: the learning rate is 0.001, the exponential decay rate is 0.9 and 0.999, the ambiguity factor is 10^{-8}, and the weight decay is 0. The loss function is a binary cross-entropy loss function. The difference is that we accumulate the binary cross-entropy loss functions of each layer and calculate the average value. In addition, in order to prove that MMS-Net can still perform well in the scenario of limited training dataset, we further reduced the training data by dividing the datasets according to the ratio of 5:5 and 2:8 for training and testing.

4.4. Comparison with state-of-the-art methods

In order to ensure the reliability of experiments and results, seven carefully selected segmentation methods (proposed in the year 2015–2023) were re-implemented by us under the same experimental configuration on all five datasets for a fair comparison, including the classic U-Net [30], transformer-based SwinU-Net [31], multi-layer perceptron-based UNext [32], multi-scale-based AttentionU-Net [33], CPF-Net [34], DCSAU-Net [35], and multi-modal based CE-Net [36].

U-Net [30] is a classical encoder–decoder structure, which has the advantages of adapting to medical images, processing data scarcity, strong feature extraction ability, and advanced performance. Therefore, it is widely used for performance comparison in medical image segmentation. Moreover, in our proposed MMS-Net, we replaced the double layers convolution of U-Net with the SCMs module, performance improvement can validate the effectiveness of SCMs.

SwinU-Net [31] has been constructed as a pure transform segmentation structure with a symmetric encoder–decoder with skip connections. It achieves self-attention from local to global in the encoder. In the decoder, global features are upsampled to the input resolution for corresponding pixel-level segmentation prediction. UNext [32] is an image segmentation network based on multilayer perceptron (MLP). The author proposes a tokenized MLP block that effectively labels and projects convolutional features and models the representation using MLP. In order to further improve performance, the input channel is changed to focus on learning local dependencies before the data is inputted into MLP. The feature extraction methods of the proposed MMS-Net, SwinU-Net and UNext are convolution, transformer, and MLP, respectively, their comparison can help to verify the effectiveness of the feature extraction method in MMS-Net.

AttentionU-Net [33] introduces an attention mechanism to suppress irrelevant regions of the input image while highlighting the prominent features of specific local regions; The attention mechanism can also establish a stronger connection between the encoder and decoder, enabling the decoder to better utilize the contextual information in the encoder. Its multi-scale feature processing has been successfully applied in medical image segmentation tasks. Context pyramid fusion Network (CPF-Net) [34] integrates global multi-scale features by combining two pyramid modules. The global pyramid guidance module aims to provide different levels of global multi-scale features for the decoder by reconstructing skip connections. The scale perception pyramid fusion module aims to dynamically fuse multi-scale features from high-level features. This method has achieved excellent results in skin lesion segmentation, retinal linear lesion segmentation, multi-class segmentation of thoracic organs at risk, and retinal edema lesions. DCSAU-Net [35] uses optimized deep separable convolution to collect shallow-level features. At the same time, it proposes lightweight multi-scale segmented attention blocks for deep feature extraction, and its compact model achieves significant performance improvements on complex images. Our MMS-Net is proposed from the perspective of facilitating multi-scale segmentation, and AttentionU-Net, CPF-Net, and DCSAU-Net all benefit from multi-scale theory, we compared them to highlight the multi-scale segmentation performance.

CE-Net [36] is a context encoder network that can capture higher-level information and preserve spatial information. It aims to solve the problem of continuous pooling and cross-convolution operations leading to the loss of some spatial information. This method has achieved excellent results in multi-modality datasets, e.g., disc segmentation, retinal vessel detection, CT lung segmentation, cell contour segmentation, and retinal optical coherence tomography layer segmentation. Our proposed MMS-Net is dedicated to multi-modality targets, so CE-Net was also selected for comparison.

Table 2
Performance comparison of different networks on the ISIC2017 dataset.

Methods	ACC	SEN	SPE	IoU	DSC
U-Net [30] (2015)	89.39	74.14	97.83	64.14	74.59
SwinU-Net [31] (2023)	93.59	81.62	97.93	76.53	84.98
UNext [32] (2022)	92.74	79.04	**98.46**	74.41	83.22
AttentionU-Net [33] (2019)	89.07	72.69	97.72	63.90	74.29
CPF-Net [34] (2020)	92.58	81.00	97.90	74.76	83.63
DCSAU-Net [35] (2023)	89.88	77.04	97.04	66.73	76.08
CE-Net [36] (2019)	93.02	81.21	97.42	75.19	83.83
MMS-Net (Proposed)	**95.40**	**87.50**	97.20	**77.90**	**87.60**

Table 3
Performance comparison of different networks on the LUNA16 dataset.

Methods	ACC	SEN	SPE	IoU	DSC
U-Net [30] (2015)	98.98	97.97	99.45	95.95	97.92
SwinU-Net [31] (2023)	99.05	98.35	99.43	96.27	98.09
UNext [32] (2022)	98.93	97.84	99.42	95.73	97.81
AttentionU-Net [33] (2019)	98.95	97.78	99.45	95.83	97.86
CPF-Net [34] (2020)	98.98	97.87	99.46	95.99	97.95
DCSAU-Net [35] (2023)	99.08	98.08	99.54	96.43	98.18
CE-Net [36] (2019)	98.99	97.88	99.46	95.97	97.93
MMS-Net (Proposed)	**99.10**	**99.00**	**99.56**	**96.44**	**98.40**

4.5. Performance evaluation

We used five metrics to quantitatively evaluate the agreement of segmentation result and ground truth, namely accuracy (ACC), sensitivity (SEN), specificity (SPE), intersection over union (IoU), and dice similarity coefficient (DSC).

$$\begin{cases} ACC = \frac{TP+TN}{TP+TN+FP+FN} \\ SEN = \frac{TP}{TP+FN} \\ SPE = \frac{TN}{TN+FP} \\ IoU = \frac{TP}{TP+FN+FP} \\ DSC = \frac{2*TP}{(TP+FN)+(TP+FP)} \end{cases} \tag{5}$$

where TP means true positive, TN means true negative, FP means false positive, FN means false negative.

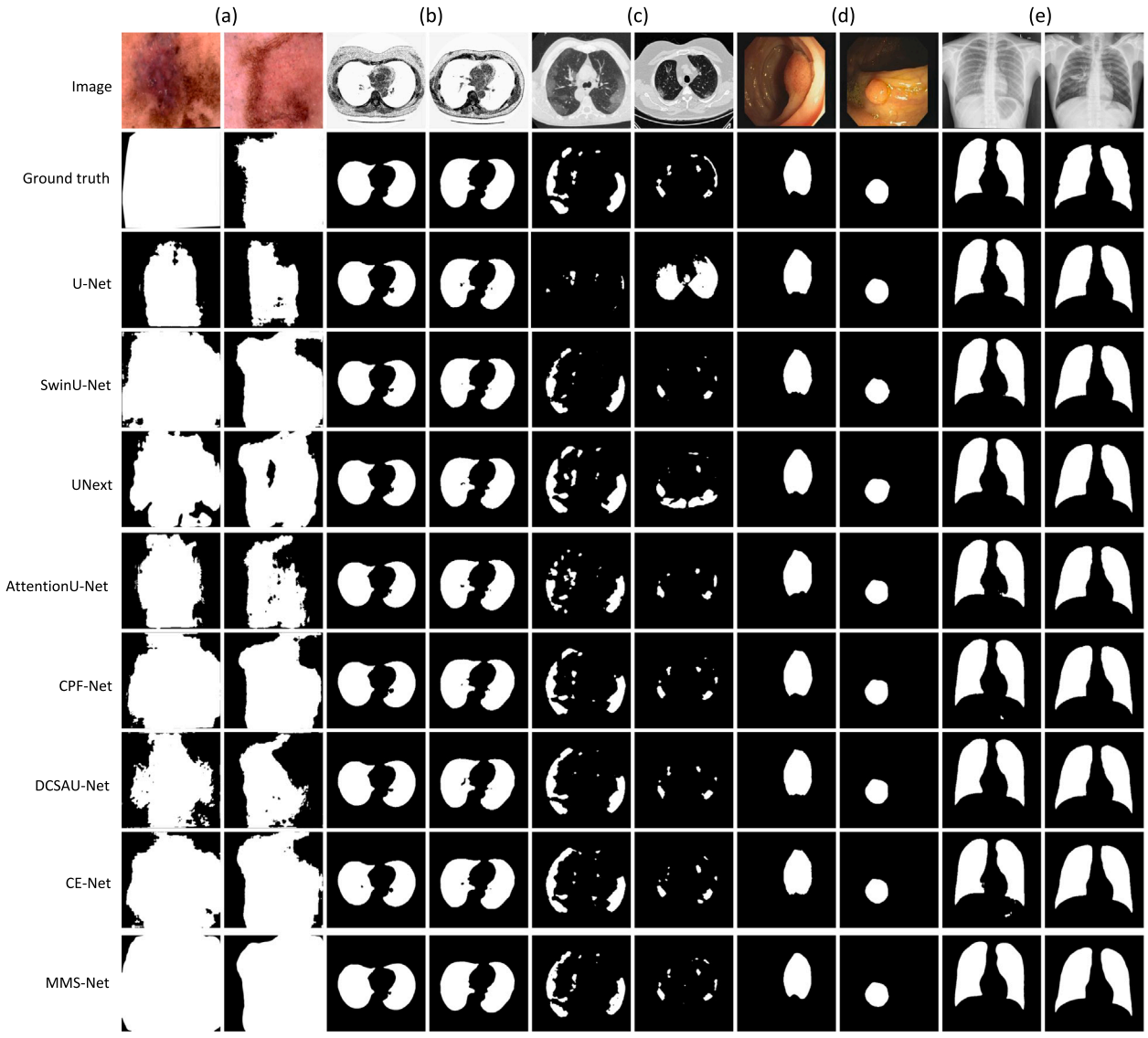


Fig. 4. Qualitative visual comparison of segmentation results for representative cases by different models in datasets of (a) ISIC2017, (b) LUNA16, (c) COVID19, (d) CVC-ClinicDB, and (e) ShenZhen.

Table 4
Performance comparison of different networks on the COVID-19 dataset.

Methods	ACC	SEN	SPE	IoU	DSC
U-Net [30] (2015)	88.58	65.64	92.59	35.39	47.60
SwinU-Net [31] (2023)	96.05	66.51	98.96	58.52	73.22
UNext [32] (2022)	95.57	84.35	96.23	58.29	72.22
AttentionU-Net [33] (2019)	96.10	69.81	98.53	58.95	73.40
CPF-Net [34] (2020)	96.44	75.50	98.50	63.05	76.75
DCSAU-Net [35] (2023)	96.85	73.91	**98.99**	**65.21**	78.38
CE-Net [36] (2019)	96.29	75.93	98.23	61.26	75.12
MMS-Net (Proposed)	**97.20**	**87.30**	97.80	64.90	**78.70**

Table 5
Performance comparison of different networks on the CVC-ClinicDB dataset.

Methods	ACC	SEN	SPE	IoU	DSC
U-Net [30] (2015)	98.52	88.34	99.44	82.29	88.63
SwinU-Net [31] (2023)	98.91	90.40	99.39	85.18	90.50
UNext [32] (2022)	98.64	87.59	99.29	81.07	87.57
AttentionU-Net [33] (2019)	98.53	90.18	99.25	83.09	89.15
CPF-Net [34] (2020)	98.80	90.44	99.36	84.60	90.12
DCSAU-Net [35] (2023)	98.90	91.02	99.45	85.04	90.52
CE-Net [36] (2019)	98.75	89.58	99.34	84.18	89.84
MMS-Net (Proposed)	**99.18**	**91.71**	**99.50**	**87.30**	**93.20**

5. Results and discussion

5.1. Performance on ISIC2017 dataset

As shown in Table 2, when 80% of the dataset was used for training. Our MMS-Net ranks first among all seven compared methods in four metrics, with ACC, SEN, IoU, and DSC of 95.40%, 87.50%, 77.90%, and 87.60%, respectively. Compared to the second-placed method SwinU-Net, the ACC, SEN, IoU, and DSC increased by 1.81%, 5.88%, 1.37% and 2.62%, respectively.

Two representative cases were displayed for visual comparison in Fig. 4(a). The image size of the ISIC2017 dataset is relatively larger than other datasets, lesions are also concentrated in large areas of the images. The segmentation results of MMS-Net are most close to the ground truth, while all the other seven comparative methods appear to be under segmentation. This is probably due to multiple local multi-scale contexts SCMs are connected, and the extracted multi-scale features are enhanced multiple times, so large-scale dermatosis images can be well segmented. In addition, MMS-Net can better distinguish foreground

and background, while other comparative methods mistakenly predict foreground as background due to insufficient extraction of large-scale features.

5.2. Performance on LUNA16 dataset

In contrast to the large image size in ISIC2017 dataset, images in LUNA16 dataset only have a fixed domination of 512 * 512. Due to the high proportion of the lung field to the whole image area in LUNA16, this segmentation task becomes easier. As listed in Table 3, the improvement of MMS-Net compared to other models is relatively small though it achieved the highest performance with ACC, SEN, SPE, IoU, and DSC of 99.10%, 99.00%, 99.56%, 96.44%, and 98.40%, respectively.

From the visual comparison results as shown in Fig. 4(b), it is easy to notice that all methods produced similar segmentation results that approached the ground truth. This dataset was previously processed when accessible, intensity distribution/contrast is quite different from the commonly used lung window in CT images, thus limiting the superiority of our proposed MMS-Net.

5.3. Performance on COVID-19 CT dataset

COVID-19 CT dataset is a small dataset with only 100 images, besides, the pneumonia area usually presents as dispersive small-scale lesions, which results in difficulty in segmentation.

As shown in Table 4, our MMS-Net yielded the highest ACC, SEN, and DSC of 97.20%, 87.30%, and 78.70%, respectively, significantly higher than the baseline U-Net with ACC, SEN, and DSC of 88.85%, 65.64%, and 47.60%, respectively. In contrast, the SPE (97.80% vs. 98.99%) and IoU (64.90% vs. 65.21%) are slightly lower than that of DCSAU-Net model with a decrease of 1.19% and 0.31%, respectively.

We also give a few examples for visual comparison in Fig. 4(c). The lesion area of COVID-19 is small and scattered, making it difficult to segment compared to other datasets. U-Net and UNext both showed over-segmentation when dealing with blurry and irregular boundaries. The local multi-scale context of multiple SCMs and the global multi-scale of FPN enable MMS-Net to have good multi-scale segmentation capabilities. The progressively decreasing SCMs can prevent MMS-Net from losing semantic features and location information when detecting small targets.

5.4. Performance on CVC-ClinicDB dataset

We also tested the generalization and performance of MMS-Net in colonoscopy images for polyp segmentation (Table 5). Among all the models, our MMS-Net performed the best in all five evaluation metrics, compared with the second-placed method DCSAU-Net, the ACC, SEN, SPE, IoU, and DSC increased by 0.28% (99.18% vs. 98.90%), 0.69% (91.71% vs. 91.02%), 0.05% (99.50% vs. 99.45%), 2.26% (87.30% vs. 85.04%), and 2.68% (93.20% vs. 90.52%).

Visual comparison of segmentation results can be seen in Fig. 4(d). The overall size of the CVC-ClinicDB image is the smallest. However, compared to the ISIC2017, LUNA16, and COVID-19 datasets, the lesion area of CVC-ClinicDB is relatively large and concentrated. The multi-level multi-scale method of MMS-Net can effectively cope with changes in image size and effectively segment concentrated lesion areas.

5.5. Performance on ShenZhen dataset

The effectiveness of MMS-Net was confirmed by dermoscopy and CT images. In order to further verify the generalization of MMS-Net, we conducted experiments on chest X-ray images to segment the whole lung field. The ShenZhen dataset has a large size, it ranges from [1233, 3001] in length to [989, 3001] in width. The results are shown in Table 6, when the model was trained with 80% of data, our MMS-Net ranks first in all five metrics. Compared to the second place in various metrics, the ACC, SEN, SPE, IoU, and DSC increased by 0.78% (98.80% vs. 98.02%), 2.46% (97.90% vs. 95.44%), 0.07% (99.10% vs. 99.03%), 2.79% (95.30% vs. 92.51%), and 1.57% (97.60% vs. 96.03%), respectively.

Table 6
Performance comparison of different networks on the ShenZhen dataset.

Methods	ACC	SEN	SPE	IoU	DSC
U-Net [30] (2015)	97.92	95.34	98.88	92.16	95.83
SwinU-Net [31] (2023)	97.91	95.28	98.92	92.04	95.74
UNext [32] (2022)	97.93	95.03	99.00	92.13	95.80
AttentionU-Net [33] (2019)	97.97	95.25	98.96	92.25	95.88
CPF-Net [34] (2020)	97.90	95.32	98.88	92.07	95.76
DCSAU-Net [35] (2023)	98.02	95.35	99.03	92.51	96.03
CE-Net [36] (2019)	97.97	95.44	98.94	92.32	95.92
MMS-Net (Proposed)	**98.80**	**97.90**	**99.10**	**95.30**	**97.60**

The visual comparison results are shown in Fig. 4(e). The ShenZhen dataset has a large image size, and the lesion area is large and concentrated. Unlike ISIC2017, the ShenZhen dataset typically has two lung regions to segment, while ISIC2017 typically only has one lesion region to segment. The superposition of multiple SCMs and FPN allows for good segmentation of large lesion areas.

5.6. Ablation experiment

In the proposed MMS-Net, SCMs and PFN are two essential components that contribute to enhanced performance. To verify their effectiveness, we conducted two ablation experiments. Firstly, we compared MMS-Net (SCMs) with U^2-Net (RSU) to verify its effectiveness. In addition, we also evaluated the superiority of FPN, i.e., connected by En^0/De^0, En^1/De^1, En^2/De^2, and En^3/De^3 in sequence, two alternative connection strategies were implemented for comparison, i.e., FPN was constructed by using four En^0/De^0 or En^3/De^3 modules, among which, when using four identical En^0/De^0 modules, it means four four-layer encoder–decoder modules are connected to each other through the network of feature pyramid.

The ablation experimental results are summarized in Table 7. When applying RSU module (i.e., U^2-Net) instead of SCMs module, model performance was reduced in all five investigated datasets, with DSC decreasing from 87.6%, 98.4%, 78.7%, 93.2% and 97.6% to 86.3%, 97.7%, 77.9%, 92.4% and 95.9% for dataset of ISIC2017, LUNA16, COVID-19, CVC-ClinicDB and ShenZhen, respectively. This can be attributed to that SCMs can extract features of different sizes through down- or up-sample operation, and fuse the features between the adjacent stages by skip connections, while RSU ignores multi-scale feature fusion. When implementing different FPN connection strategies, the model performance also experienced varying degrees of degradation, especially in datasets of ISIC2017 and CVC-ClinicDB, even more than a 5% decrease was observed in ACC, SEN, IoU, and DSC. The most declining performance occurred in CVC-ClinicDB dataset, with IoU decreased from 87.3% to 76.6% when FPN was constructed by using four En^0/De^0 modules. The visual comparison of the ablation experiment is elaborated in Fig. 5. The visualization results also demonstrated that MMS-Net has good segmentation ability in datasets of different modalities, image sizes, and data scales.

5.7. Low dependency on training dataset scale

The increase in model layers needs to be matched with larger size of the training dataset, otherwise, it will lead to gradient problems and ultimately result in poor testing results. In order to ensure consistency of the data partitioning with the comparison methods, we used 80% for training and validation in Sections 5.1–5.5. In this section, to verify the low dependency of MMS-Net on the training dataset scale, training

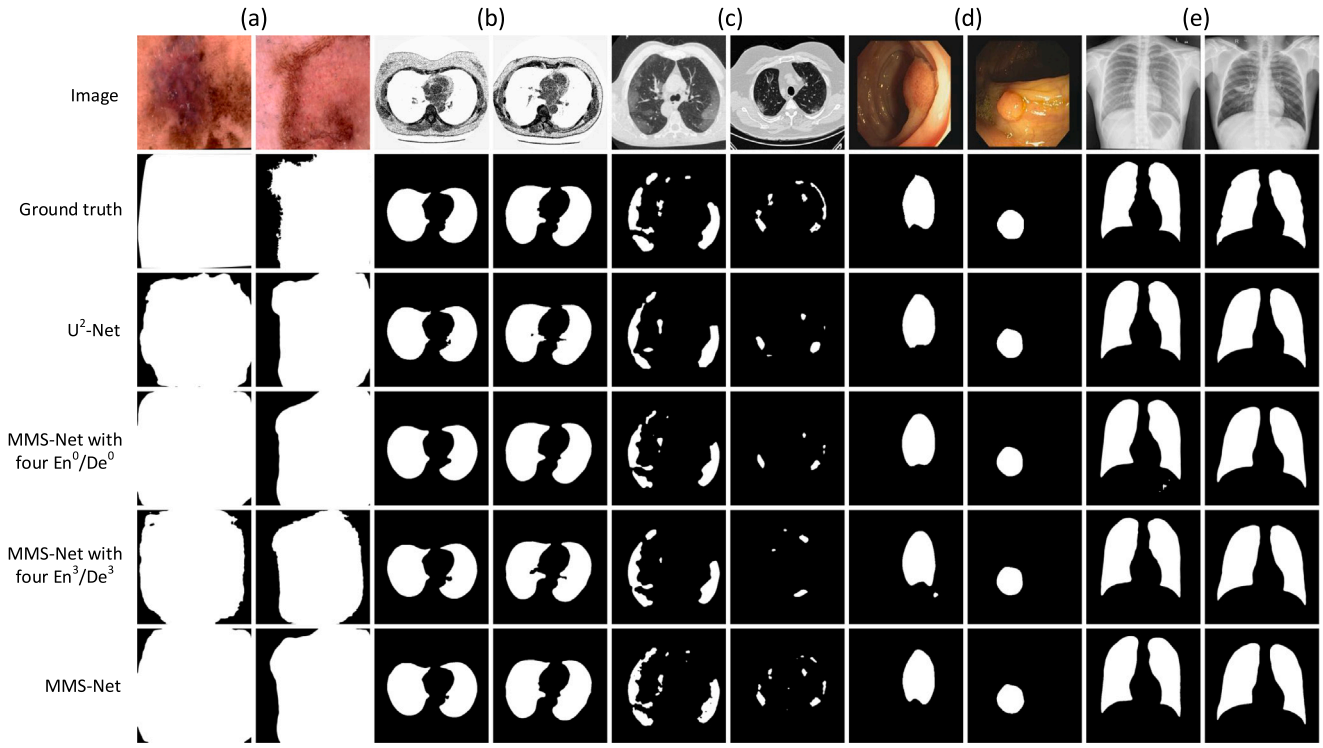


Fig. 5. Visual comparison of ablation experiment between the ground truth and the predicted result of representative cases in datasets of (a) ISIC2017, (b) LUNA16, (c) COVID-19, (d) CVC-ClinicDB, and (e) ShenZhen.

Table 7
Ablation experiment.

DataSet	Models	ACC	SEN	SPE	IoU	DSC
ISIC2017	U^2-Net [37]	94.6	80.3	**98.5**	76.0	86.3
	MMS-Net with four En^0/De^0	90.0	79.6	95.1	72.3	83.9
	MMS-Net with four En^3/De^3	91.9	85.0	95.0	76.2	86.5
	MMS-Net(Proposed)	**95.4**	**87.5**	97.2	**77.9**	**87.6**
LUNA16	U^2-Net [37]	98.9	97.7	99.3	95.4	97.7
	MMS-Net with four En^0/De^0	99.0	97.3	99.5	95.8	97.8
	MMS-Net with four En^3/De^3	99.1	98.6	99.2	95.9	97.9
	MMS-Net (Proposed)	**99.1**	**99.0**	**99.6**	**96.4**	**98.4**
COVID-19	U^2-Net [37]	97.1	88.7	97.6	63.9	77.9
	MMS-Net with four En^0/De^0	96.1	92.1	96.4	62.3	76.8
	MMS-Net with four En^3/De^3	96.0	**93.4**	96.2	61.3	76.0
	MMS-Net(Proposed)	**97.2**	87.3	**97.8**	**64.9**	**78.7**
CVC-ClinicDB	U^2-Net [37]	98.8	**93.2**	99.3	85.9	92.4
	MMS-Net with four En^0/De^0	97.8	85.2	99.0	76.6	86.7
	MMS-Net with four En^3/De^3	98.4	90.0	99.2	82.5	90.4
	MMS-Net(Proposed)	**99.2**	91.7	**99.5**	**87.3**	**93.2**
ShenZhen	U^2-Net [37]	98.0	97.0	98.3	92.2	95.9
	MMS-Net with four En^0/De^0	97.9	96.4	98.5	92.1	95.9
	MMS-Net with four En^3/De^3	98.0	96.8	98.4	92.2	96.0
	MMS-Net (Proposed)	**98.8**	**97.9**	**99.1**	**95.3**	**97.6**

Table 8
Performance comparison of MMS-Net trained with different proportions of training dataset.

DataSet	Proportion of training dataset	ACC	SEN	SPE	IoU	DSC
ISIC2017	80%	95.4	**87.5**	97.2	**77.9**	**87.6**
	50%	95.4	82.1	**98.4**	76.7	86.8
	20%	**95.6**	85.3	97.8	77.3	87.2
LUNA16	80%	**99.1**	**99.0**	**99.6**	**96.4**	**98.4**
	50%	98.9	97.7	98.2	95.2	97.3
	20%	98.9	97.8	98.3	95.8	97.5
COVID-19	80%	**97.2**	**87.3**	97.8	**64.9**	**78.7**
	50%	96.3	79.0	97.8	62.8	77.1
	20%	96.4	71.2	**98.6**	60.7	75.6
CVC-ClinicDB	80%	**99.2**	**91.7**	**99.5**	**87.3**	**93.2**
	50%	98.1	82.2	98.9	85.5	92.2
	20%	97.8	79.9	97.3	73.5	84.7
ShenZhen	80%	**98.8**	**97.9**	**99.1**	**95.3**	**97.6**
	50%	98.1	96.7	98.6	92.5	96.1
	20%	97.9	96.3	98.5	92.1	95.9

datasets were further decreased to 50% and 20%. We assume that the segmentation difficulty of different samples in the test set is consistent, meaning that the partitioning of the test set will not affect the final performance evaluation.

As listed in Table 8, in the dataset of ISIC2017, when 50% of the dataset was used for training and validation, DSC only decreased by 0.8%. Furthermore, when using a 20% dataset for training and validation, DSC even improved by 0.4% compared to using a 50% dataset for training and validation. The above results indicate that the performance of MMS-Net is less affected by the size of the dataset.

The experimental results of LUNA16, COVID-19, and ShenZhen datasets are consistent with the results of ISIC2017. When the size of the training and validation sets of the dataset decreased, the decrease of DSC is within an acceptable range. Among them, due to the overall small sample size of the dataset COVID-19, the most significant decrease in DSC was 3.1%.

Among different partition ratios of CVC-ClinicDB, DSC was only decreased by 1% when using 50% data for training compared with using 80%. However, due to the fact that the original image of CVC-ClinicDB is only 384 * 288, DSC of 84.7% was only obtained when 20% is used for training.

In summary, comparable or slightly lower performance can be well maintained by MMS-Net even with limited training dataset, MMS-Net is less constrained by the training dataset scale.

5.8. Ability to segment small lesions

The essence of our proposed MMS-Net is its ability to extract multi-scale features and thus enable to segmentation of both large and small

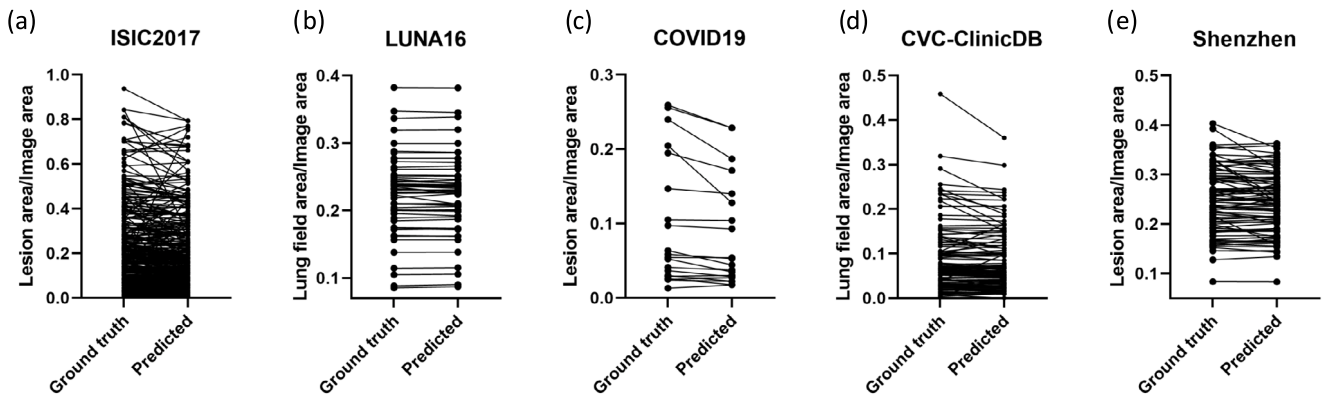


Fig. 6. The proportion of lesion/lung field area to the whole image area calculated from the ground truth and predicted masks in datasets of (a) ISIC2017, (b) LUNA16, (c) COVID-19, (d) CVC-ClinicDB, and (e) ShenZhen.

lesions. To evaluate the ability of MMS-Net to segment small lesions, we compared the predicted lesion size with the ground truth. Since the absolute lesion areas among different datasets showed great variation, we calculated the proportion of lesion/lung field area to the whole image area as an alternative. Fig. 6 showed the proportion change between the ground truth and predicted masks. The proposed model holds great ability to accurately segment small lesions. The LUNA16 and ShenZhen datasets are CT and X-ray images respectively, which are used to segment the whole lung field, since the shape, size, and other information of lung field can be correctly modeled, good segmentation performance was obtained on these two datasets. However, in ISIC2017, COVID-19, and CVC-ClinicDB datasets, the shape, size, color, and texture information of lesions are greatly varied, thus inferior performance was achieved.

We give three representative cases in each dataset for further visualizing the difference between the ground truth and the segmentation results as shown in Fig. 7. In ISIC2017 dataset, over-segmentation and under-segmentation occurred in cases with blurry and irregular boundaries regardless of their large size of lesion, good performance was observed in lesions with high contrast despite the small size. Best segmentation was also shown. In COVID-19 dataset, the model can correctly detect isolated and small-size lesions, while for pulmonary fibrosis, the model failed to segment regions with low intensity. In CVC-ClinicDB dataset, illumination greatly affected the segmentation performance. In ShenZhen dataset, satisfactory performance was achieved although smaller lung field was obtained due to the interference of ribs or lung lesions.

5.9. Limitations

This study evaluated our proposed segmentation method MMS-Net in five datasets (Section 4.1), including dermatoscopy images for melanoma segmentation, lung CT images for lung field segmentation, lung CT images for COVID-19 pneumonia segmentation, colonoscope images for polyp segmentation and chest X-ray images for lung field segmentation. These five datasets covered different modalities, lesions/organs with a wide range of image sizes (288–6728), and a different number of samples (100–2000) that were concerned by different clinical departments, we believe they are representative enough in this study though there are even more modalities and more specific applications in the area of medical image segmentation, e.g., head and neck tumor segmentation in PET images [38], breast tumor segmentation in MRI images [39], etc. Future works can be extended to wider clinical applications. Besides, all the formats of images are .png or .jpg as only these formats were available from the original links, many previous publications [9,10,36] have also adopted .png or .jpg format of CT or X-ray images for their segmentation model construction. DICOM format with higher bits may convey the most original information for CT and X-ray images, future works include seeking cooperation from hospitals and focusing on DICOM format for radiological image segmentation.

6. Conclusion

This study proposed a MMS-Net to establish the relationship among the upper-, the lower-, and the same layers through feature pyramid network (FPN), and extract the global multi-scale information through up-sample and down-sample in SCMs. We superimposed SCMs with different layers, which can deepen the layers of the network and increase the nonlinear fitting ability of the model. We combined the global multi-scale feature and local multi-scale context into multi-level multi-scale feature, which ensures the segmentation ability of MMS-Net for lesions with different sizes, especially for the segmentation of scattered small infected lesions. MMS-Net has been proven to be effective in dermoscopy, colonoscopy, CT, and X-ray experiments.

Funding

This work was funded by the National Natural Science Foundation of China (62062067 and 12026601), the Program of Yunnan Key Laboratory of Intelligent Systems and Computing (202205AG070003), the Guangdong Basic and Applied Basic Research Foundation (2020A1515 110683), and the Open Project Program of Yunnan Key Laboratory of Intelligent Systems and Computing (ISC23Y02).

CRediT authorship contribution statement

Chang Zhao: Conceptualization, Methodology, Software, Validation, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization. **Wenbing Lv:** Conceptualization, Formal analysis, Resources, Data curation, Writing – original draft, Writing – review & editing, Supervision, Project administration, Funding acquisition. **Xiang Zhang:** Conceptualization, Resources, Data curation, Writing – original draft, Writing – review & editing. **Zimin Yu:** Software. **Shunfang Wang:** Conceptualization, Methodology, Resources, Writing – review & editing, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

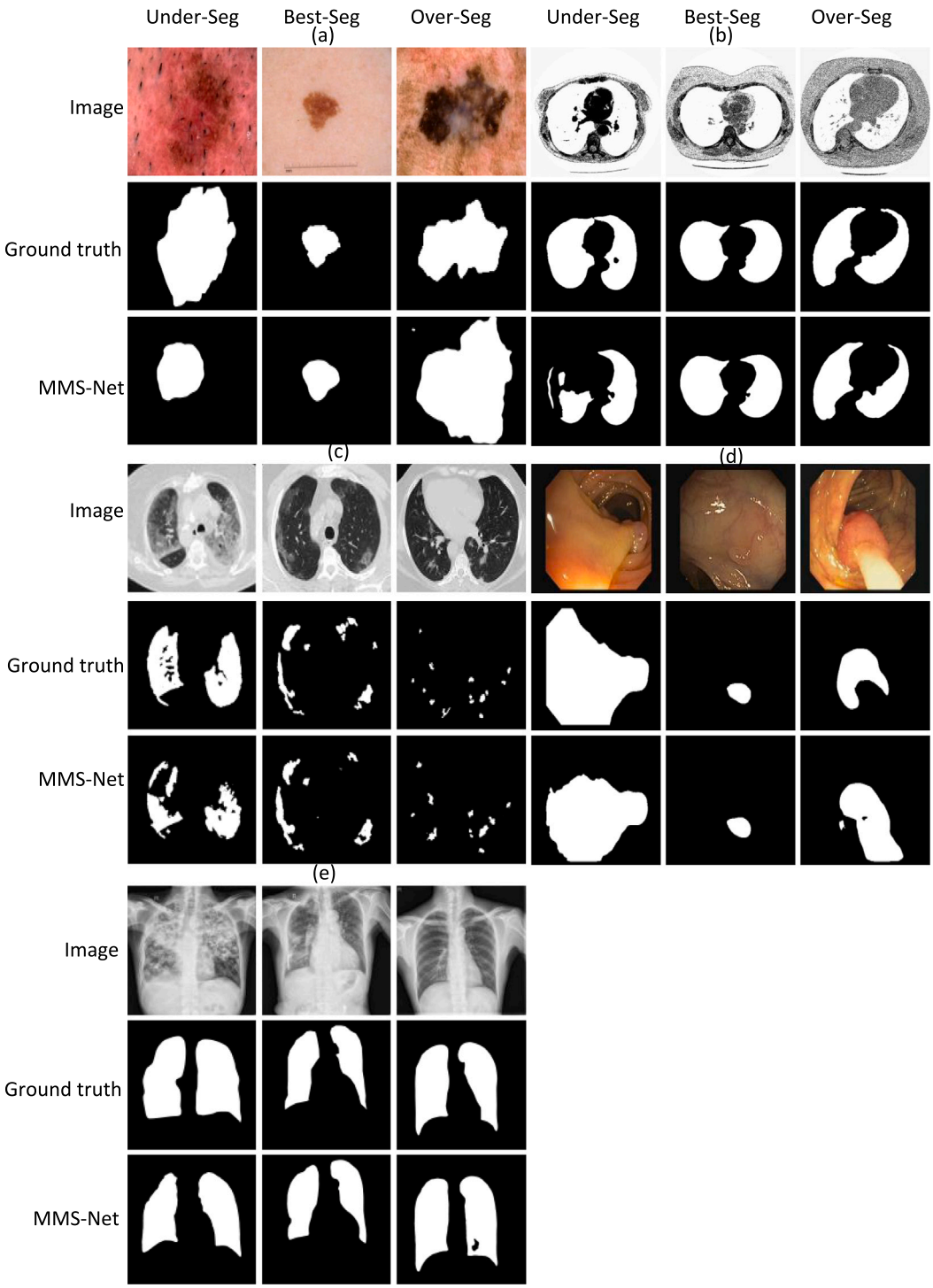


Fig. 7. Visual comparison between the ground truth and the predicted result of representative cases in datasets of (a) ISIC2017, (b) LUNA16, (c) COVID-19, (d) CVC-ClinicDB, and (e) ShenZhen.

References

[1] M. Astaraki, C. Wang, Prior-aware autoencoders for lung pathology segmentation, Med. Image Anal. 80 (2022) 102491.

[2] L. Li, H. Zhao, H. Wang, W. Li, S. Zheng, Automatic abdominal segmentation using novel 3D self-adjustable organ aware deep network in CT images, Biomed. Signal Process. Control 84 (2023) 104691.

[3] X. Zhao, L. Zhang, H. Lu, Automatic polyp segmentation via multi-scale subtraction network, in: International Conference on Medical Image Computing and Computer-Assisted Intervention, Springer, 2021, pp. 120–130.

[4] C. Yang, X. Guo, Z. Chen, Y. Yuan, Source free domain adaptation for medical image segmentation with fourier style mining, Med. Image Anal. 79 (2022) 102457.

[5] Y. Zhou, C. Koyuncu, C. Lu, R. Grobholz, I. Katz, A. Madabhushi, A. Janowczyk, Multi-site cross-organ calibrated deep learning (MuSClD): Automated diagnosis of non-melanoma skin cancer, Med. Image Anal. 84 (2023) 102702.

[6] R. Wang, S. Chen, C. Ji, J. Fan, Y. Li, Boundary-aware context neural network for medical image segmentation, Med. Image Anal. 78 (2022) 102395.

[7] E. Çallı, E. Sogancioglu, B. van Ginneken, K.G. van Leeuwen, K. Murphy, Deep learning for chest X-ray analysis: A survey, Med. Image Anal. 72 (2021) 102125.

[8] A. Iqbal, M. Usman, Z. Ahmed, Tuberculosis chest X-ray detection using CNN-based hybrid segmentation and classification approach, Biomed. Signal Process. Control 84 (2023) 104667.
[9] National Institutes of Health, et al., NIH clinical center provides one of the largest publicly available chest X-ray datasets to scientific community, 2017.
[10] S. Jaeger, A. Karargyris, S. Candemir, L. Folio, J. Siegelman, F. Callaghan, Z. Xue, K. Palaniappan, R.K. Singh, S. Antani, et al., Automatic tuberculosis screening using chest radiographs, IEEE Trans. Med. Imaging 33 (2) (2013) 233–245.
[11] F. Yu, V. Koltun, Multi-scale context aggregation by dilated convolutions, 2015, arXiv preprint arXiv:1511.07122.
[12] L.-C. Chen, G. Papandreou, F. Schroff, H. Adam, Rethinking atrous convolution for semantic image segmentation, 2017, arXiv preprint arXiv:1706.05587.
[13] T.-Y. Lin, P. Dollár, R. Girshick, K. He, B. Hariharan, S. Belongie, Feature pyramid networks for object detection, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2017, pp. 2117–2125.
[14] L.-C. Chen, G. Papandreou, I. Kokkinos, K. Murphy, A.L. Yuille, Deeplab: Semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected crfs, IEEE Trans. Pattern Anal. Mach. Intell. 40 (4) (2017) 834–848.
[15] C. Szegedy, V. Vanhoucke, S. Ioffe, J. Shlens, Z. Wojna, Rethinking the inception architecture for computer vision, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2016, pp. 2818–2826.
[16] T. Qu, X. Wang, C. Fang, L. Mao, J. Li, P. Li, J. Qu, X. Li, H. Xue, Y. Yu, et al., M3Net: A multi-scale multi-view framework for multi-phase pancreas segmentation based on cross-phase non-local attention, Med. Image Anal. 75 (2022) 102232.
[17] X. Geng, X. Liu, S. Cheng, S. Zeng, Cervical cytopathology image refocusing via multi-scale attention features and domain normalization, Med. Image Anal. 81 (2022) 102566.
[18] G. Valvano, A. Leo, S.A. Tsaftaris, Learning to segment from scribbles using multi-scale adversarial attention gates, IEEE Trans. Med. Imaging 40 (8) (2021) 1990–2001.
[19] Y. Chen, H. Zhang, Y. Wang, Y. Yang, X. Zhou, Q.J. Wu, MAMA Net: Multi-scale attention memory autoencoder network for anomaly detection, IEEE Trans. Med. Imaging 40 (3) (2020) 1032–1041.
[20] G. Wang, S. Zhai, G. Lasio, B. Zhang, B. Yi, S. Chen, T.J. Macvittie, D. Metaxas, J. Zhou, S. Zhang, Semi-supervised segmentation of radiation-induced pulmonary fibrosis from lung CT scans with multi-scale guided dense attention, IEEE Trans. Med. Imaging 41 (3) (2021) 531–542.
[21] Z.-L. Ni, X.-H. Zhou, G.-A. Wang, W.-Q. Yue, Z. Li, G.-B. Bian, Z.-G. Hou, SurgiNet: Pyramid attention aggregation and class-wise self-distillation for surgical instrument segmentation, Med. Image Anal. 76 (2022) 102310.
[22] C. Dong, S. Xu, D. Dai, Y. Zhang, C. Zhang, Z. Li, A novel multi-attention, multi-scale 3D deep network for coronary artery segmentation, Med. Image Anal. 85 (2023) 102745.
[23] S. Xie, Z. Tu, Holistically-nested edge detection, in: Proceedings of the IEEE International Conference on Computer Vision, 2015, pp. 1395–1403.
[24] R. Girshick, Fast r-cnn, in: Proceedings of the IEEE International Conference on Computer Vision, 2015, pp. 1440–1448.
[25] N. Codella, D. Gutman, M.E. Celebi, B. Helba, M.A. Marchetti, S.W. Dusza, A. Kalloo, K. Liopyris, N. Mishra, H. Kittler, et al., Skin lesion analysis toward melanoma detection: A challenge at the 2017 International Symposium on Biomedical Imaging (ISBI), hosted by the International Skin Imaging Collaboration (ISIC), IEEE Trans. Med. Imaging 38 (9) (2019) 2103–2118.
[26] A.A. Setio, A. Traverso, T. de Bel, M.S. Berens, C. van den Bogaard, P. Cerello, H. Chen, Q. Dou, M.E. Fantacci, B. Geurts, et al., Validation, comparison, and combination of algorithms for automatic detection of pulmonary nodules in computed tomography images: the LUNA16 challenge, Med. Image Anal. 42 (2017) 1–13.
[27] Jenssen, COVID-19 CT segmentation dataset, 2020, Available online: https://medicalsegmentation.com/covid19/.
[28] R. Cossio, M. Villegas, J.R. Orozco-Arroyave, C.G. Castellanos-Domínguez, T. Arias-Vergara, G. Rueda-Morales, et al., CVC-ClinicDB: A new handwritten text database including personal health cards, ICDAR 1 (2015) 1211–1215.
[29] L. Wang, D.W. Wong, J. Liu, P. Shi, X. Zhou, C. Wang, X. Huang, S. Zhang, A benchmark for comparison of dental radiography analysis algorithms, Med. Image Anal. 36 (2017) 61–76.
[30] O. Ronneberger, P. Fischer, T. Brox, U-net: Convolutional networks for biomedical image segmentation, in: Medical Image Computing and Computer-Assisted Intervention–MICCAI 2015: 18th International Conference, Munich, Germany, October 5-9, 2015, Proceedings, Part III 18, Springer, 2015, pp. 234–241.
[31] H. Cao, Y. Wang, J. Chen, D. Jiang, X. Zhang, Q. Tian, M. Wang, Swin-unet: Unet-like pure transformer for medical image segmentation, in: European Conference on Computer Vision, Springer, 2022, pp. 205–218.
[32] J.M.J. Valanarasu, V.M. Patel, Unext: Mlp-based rapid medical image segmentation network, in: Medical Image Computing and Computer Assisted Intervention–MICCAI 2022: 25th International Conference, Singapore, September 18–22, 2022, Proceedings, Part V, Springer, 2022, pp. 23–33.
[33] J. Schlemper, O. Oktay, M. Schaap, M. Heinrich, B. Kainz, B. Glocker, D. Rueckert, Attention gated networks: Learning to leverage salient regions in medical images, Med. Image Anal. 53 (2019) 197–207.
[34] S. Feng, H. Zhao, F. Shi, X. Cheng, M. Wang, Y. Ma, D. Xiang, W. Zhu, X. Chen, CPFNet: Context pyramid fusion network for medical image segmentation, IEEE Trans. Med. Imaging 39 (10) (2020) 3008–3018.
[35] Q. Xu, Z. Ma, H. Na, W. Duan, DCSAU-Net: A deeper and more compact split-attention U-net for medical image segmentation, Comput. Biol. Med. 154 (2023) 106626.
[36] Z. Gu, J. Cheng, H. Fu, K. Zhou, H. Hao, Y. Zhao, T. Zhang, S. Gao, J. Liu, CE-Net: Context encoder network for 2d medical image segmentation, IEEE Trans. Med. Imaging 38 (10) (2019) 2281–2292.
[37] X. Qin, Z. Zhang, C. Huang, M. Dehghan, O.R. Zaiane, M. Jagersand, U2-Net: Going deeper with nested U-structure for salient object detection, Pattern Recognit. 106 (2020) 107404.
[38] V. Andrearczyk, V. Oreiller, M. Abobakr, A. Akhavanallaf, P. Balermpas, S. Boughdad, L. Capriotti, J. Castelli, C. Cheze Le Rest, P. Decazes, R. Correia, D. El-Habashy, H. Elhalawani, C.D. Fuller, M. Jreige, Y. Khamis, A. La Greca, A. Mohamed, M. Naser, J.O. Prior, S. Ruan, S. Tanadini-Lang, O. Tankyevych, Y. Salimi, M. Vallières, P. Vera, D. Visvikis, K. Wahid, H. Zaidi, M. Hatt, A. Depeursinge, Overview of the HECKTOR challenge at MICCAI 2022: Automatic head and neck tumor segmentation and outcome prediction in PET/CT, in: V. Andrearczyk, V. Oreiller, M. Hatt, A. Depeursinge (Eds.), Head and Neck Tumor Segmentation and Outcome Prediction, Springer Nature Switzerland, Cham, 2023, pp. 1–30.
[39] S. Wang, K. Sun, L. Wang, L. Qu, F. Yan, Q. Wang, D. Shen, Breast tumor segmentation in DCE-MRI with tumor sensitive synthesis, IEEE Trans. Neural Netw. Learn. Syst. (2021) 1–12.